

Environmental Station

Schedule observations; preserve evidence; present health

The LCD explains the record. The RGB cadence proves the station is alive.

Lesson	015 — integration project	Time	180–240 minutes
Prerequisites	Lessons 001–014	Board	Arduino Mega 2560
Evidence	LCD age/status, D5–D7 health LED, TP13 DATA	Status	host verified; physical acceptance open
Safety	E1; USB-powered low-current circuit only		

Safety checkpoint. This is a learning instrument, not a calibrated weather station, alarm, or life-safety monitor. Disconnect USB before wiring. Identify the exact DHT11 part or module before adding its DATA pull-up. Every RGB channel needs its own resistor. Check LCD pin numbering against its datasheet. Stop for heat, odor, resets, an out-of-rail DATA voltage, a blank display after contrast adjustment, or disagreement between the wiring table and the actual part. Remove USB power to stop the experiment.

1 Mission

Compose the previous two component boundaries into one reproducible station:

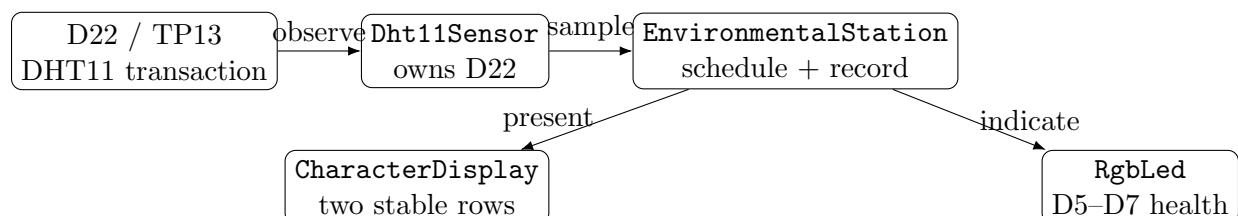
`DHT11` → `ClimateSensor` → `EnvironmentalStation` → {stable record, LCD, health LED}.

The project is deliberately strict about the evidence chain. A scheduled transaction produces a numbered record even when the sensor reports a checksum, range, timeout, or stale-data fault. The LCD presents that record; it does not decide whether the record is trustworthy. The RGB LED supplies a continuous non-Serial health path. TP13 exposes the DATA transaction itself.

By the end, you can:

- distinguish acquisition, transaction, validation, freshness, and presentation;
- explain why a caller-supplied timestamp makes the project replayable;
- predict exactly when a new record is due;
- interpret a numbered fault record without replacing it with old data;
- prove resource acquisition separately from safe shutdown;
- replay identical sensor frames and obtain identical station records.

2 Composition and ownership



`EnvironmentalStation` receives an explicit non-owning `ClimateSensor` relationship. During `initialize()` it validates the period and freshness policy, then initializes the sensor. Failure rolls the sensor back. Repeated initialization does not reacquire it. Shutdown releases the sensor, clears the schedule and record, and is idempotent; the destructor repeats this cleanup safely.

The LCD and RGB evidence are owned by the sketch because they are presentation choices, not part of the station's decision engine. Setup therefore acquires in dependency order and rolls back in reverse order:

1. sensor and station;
2. LCD;
3. RGB evidence;
4. initial blank/fault-safe presentation.

3 The deterministic record contract

Field	Meaning
<code>sample</code>	temperature, humidity, observation time, and exact sample state
<code>health</code>	starting, healthy, sensor fault, stale, or timing fault
<code>sensorStatus</code>	transport/update status without replacing sample detail
<code>recordedAt</code>	caller time at which the station recorded the observation
<code>sequence</code>	monotonically increasing record number since <code>initialize/reset</code>

The first `update(now)` is due immediately. After a record, the next deadline is `now + samplePeriod`. An early update is a no-op. A late update takes one observation and schedules from the supplied time; it does not burst through missed transactions. This prevents a blocked sensor from turning one late loop into a long catch-up loop.

`recordReady` is an event snapshot. It is true only for the update that created a record and remains readable until the next update. The record itself remains stable between deadlines. Unsigned time works across `millis()` rollover while distances remain within the half-range. Ambiguous reverse time becomes `TimingFault` and cancels the deadline until explicit reset.

```
void loop ()
{
    const adk::TimePoint      now      (millis ());
    const adk::Status         status   = station.update (now);
    const adk::EnvironmentalSnapshot evidence = station.snapshot ();

    if (evidence.recordReady)
    {
        presentRecord (evidence.record);
    }

    showHealth (evidence.record.health, status);
    display.update (now);
}
```

4 Parts, wiring, and current

Quantity	Part	Requirement
1	Mega 2560	USB powered
1	DHT11	exact bare part or documented module identified
1	16x2 HD44780 LCD	5 V-compatible module, 4-bit interface
1	10 k Ω potentiometer	LCD contrast between 5 V and GND
1	DATA pull-up	value required by identified sensor; omit when module supplies it
1	common-cathode RGB LED	three known channel pins plus common cathode
3	330 Ω resistors	one per RGB channel
1	LCD backlight resistor	only if the identified module does not include one
1	multimeter	continuity and rail checks

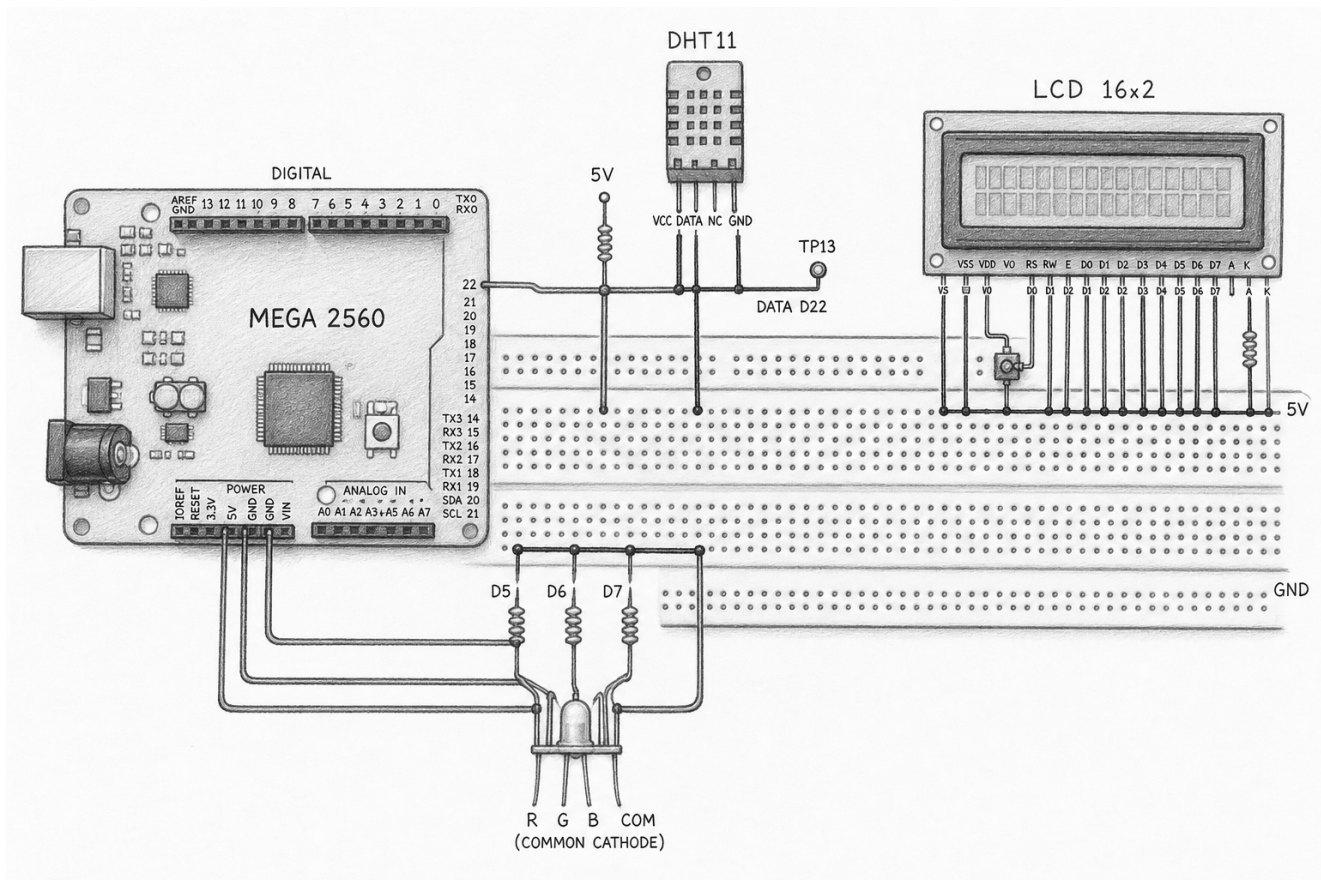


Figure 1: Pencil orientation of the Mega, DHT11, LCD, RGB health indicator, and TP13. It is not an authoritative schematic. Alternative text: Mega D22 reaches DHT11 DATA and TP13; a 16x2 LCD uses a separate six-wire 4-bit interface; D5, D6, and D7 each reach one resistor-limited RGB channel; all parts share 5 V and GND, and DHT11 NC remains open.

Endpoint	Connection	Evidence
DHT11 VDD/GND	Mega 5 V/GND	rail measurement at the sensor
DHT11 DATA	D22, pull-up if required	TP13 pulse transaction
DHT11 NC	no connection	continuity check while unpowered
LCD RS, E	D30, D31	command/data strobes
LCD D4–D7	D32–D35	4-bit data path
LCD RW	GND	write-only operation
LCD VO	contrast potentiometer wiper	readable cells after adjustment
RGB R/G/B	D5/D6/D7 through 330 Ω each	health cadence
RGB common cathode	GND	shared return

Use the actual LED forward voltages to calculate each channel current:

$$I = \frac{V_{\text{rail}} - V_f}{R}.$$

Record each value and the worst simultaneous total. Do not infer current from brightness. If the LCD module exposes a raw backlight LED, use its documented series resistance rather than assuming resistance is on the board.

5 Predict, observe, interpret

5.1 Acquisition evidence

Predict: after all resources initialize, the LCD shows a starting record and the RGB indicator begins the slow starting cadence. If any claim fails, the LCD is blanked and all RGB channels turn off.

Observe: power the checked circuit. Watch the complete startup sequence rather than a single final LED state. Confirm that the first numbered record appears only after the scheduled observation.

Interpret: the sequence proves that software acquired its objects and advanced the loop. It does not prove electrical output levels or climate accuracy.

5.2 Healthy and fault records

Predict: a valid fresh sample displays temperature/humidity and a record number while the indicator gives the healthy cadence. A timeout, checksum failure, invalid range, or stale sample remains visible as a named fault record; old valid values are not relabeled as new.

Observe: record both LCD rows and the LED cadence. With power removed, disconnect only DATA to create a controlled transport fault, recheck the wiring, then power again. Restore DATA only with power removed.

Interpret: a fault record demonstrates explicit uncertainty. A healthy record demonstrates accepted digital data, not calibration against a reference instrument.

5.3 Electrical safe state

Predict: after commanded shutdown, D22 and every owned display or RGB pin reach their documented inactive/high-impedance state. The LCD and RGB LED are off.

Observe: invoke the lesson shutdown path, then measure the named pins relative to GND. Separately remove USB power and confirm the circuit is de-energized.

Interpret: pin measurements support the safe-state claim. Merely seeing a dark LED does not: a reversed or failed LED can also appear dark.

6 Host replay and CLI workflow

The host test injects a recorded sensor rather than waiting for a room to change. Its trace covers immediate first sampling, exact deadlines, early no-ops, late updates without catch-up bursts, healthy/fault/stale recovery, initialization rollback, restart, destruction, rollover, ambiguous time, and byte-for-byte replay.

```
make test-environmental-station
make test-sanitize
make compile-Lesson015EnvironmentalStation
make size-Lesson015EnvironmentalStation
make lesson-015
make lessons-check
make upload-Lesson015EnvironmentalStation PORT=/dev/ttyACM0
make monitor PORT=/dev/ttyACM0
```

Serial monitoring is optional supporting context. The LCD, RGB cadence, and TP13 remain the required evidence paths.

The canonical Mega 2560 build uses 14,134 bytes of flash and 912 bytes of static RAM with the current Arch Arduino AVR toolchain. These are compile-size measurements, not runtime-stack or physical-current measurements.

7 Diagnosis

Observation	Likely boundary	Next check with power removed
No LCD, no RGB sequence	acquisition or supply	rail polarity, shorts, pin claims
RGB healthy, LCD blank	presentation	LCD pin 1 orientation, VO contrast, RW to GND
Fault record, TP13 quiet	sensor power/DATA	sensor identity, pull-up, D22 continuity
Fault record, TP13 active	transport/validation	timing capture, checksum/range state
Same sequence repeats	scheduling loop stalled	visible cadence and caller time path
Plausible but wrong values	environment or accuracy	compare with a documented reference; do not recalibrate by guess
LED dark after shutdown	inconclusive alone	measure D5–D7 and D22 relative to GND

8 Exercises

1. Predict the record sequence and deadlines for updates at 0, 1999, 2000, 9000, and 9001 ms with a 2000 ms period. Explain the lack of catch-up records.
2. Extend a recorded host trace with checksum failure, stale sample, and recovery. List every expected health value before running it.

3. Design two LCD rows that fit exactly 16 columns and retain sample state, age, and sequence. Explain which numeric detail you omit and why.
4. Explain why the station owns the sensor lifecycle but does not own the presentation objects.
5. Measure the RGB channel currents and compare them with the calculated values. Record rail voltage, V_f , resistor value, and uncertainty.
6. Inject a display failure in a host fake. Specify the station record that must remain preserved even though presentation failed.

9 Bench acceptance record

Item	Record
Board and USB supply	_____
DHT11 part/module source	_____
LCD part/source and backlight limit	_____
Measured 5 V rail	_____
RGB resistors and channel currents	_____
Unpowered continuity/polarity check	☐ pass ☐ stop
Startup/acquisition sequence	_____
Healthy record and cadence	_____
Controlled DATA fault record	_____
Reset/recovery observation	_____
Shutdown pin measurements	_____
Power-removed observation	_____
Instrument, date, reviewer	_____
Deviation or stop reason	_____

Release status: host verified; physical acceptance open. Compilation, host replay, generated PDF, and firmware size are not physical evidence. Do not mark this lesson hardware accepted until the completed record contains measured startup, normal, fault, reset, shutdown, and power-removed observations on the named Mega 2560 circuit.

10 Further reading

- Arduino Mega 2560 Rev3 documentation: <https://docs.arduino.cc/hardware/mega-2560/>
- DHT11 product documentation from the exact identified manufacturer or module supplier
- HD44780U controller datasheet from Hitachi/Renesas
- ADK development, testing, safety, and PDF policies in the repository